

Receptive Field Analysis

in Convolutional Neural Networks

Datasets: MNIST · CIFAR-10 · CIFAR-100

Models: SimpleCNN · DeepCNN · DepthwiseCNN

Kernel sizes: 1 · 2 · 3 · 5 · 7 · 9 · 11 · 15 · 21 · 28

Portfolio mini-study

Experiment Setup

Datasets

MNIST - 28×28 grayscale
60,000 train / 10,000 test

CIFAR-10 - 32×32 RGB, 10 classes
50,000 train / 10,000 test

CIFAR-100 - 32×32 RGB, 100 classes
50,000 train / 10,000 test

Preprocessing: ToTensor() + Normalize()

Models

SimpleCNN - 1 convolutional layer

DeepCNN - 3 convolutional layers

DepthwiseCNN - 3 depthwise-separable blocks

Global Average Pooling makes the architectures compatible with both 28×28 and 32×32 inputs.

Metrics

Test accuracy

Macro-F1 score

Trainable parameter count

Training time (s)

Theoretical convolution-stack receptive field

Kernel sizes $k \in \{1, 2, 3, 5, 7, 9, 11, 15, 21, 28\}$ · 3 datasets × 3 models × 10 kernels = 90 runs

Model Overview

SimpleCNN

$$RF = k$$

Shallow baseline with one kernel-size-dependent convolutional layer.

$\text{Conv}(k) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \rightarrow \text{GAP} \rightarrow \text{FC}(32 \rightarrow 64) \rightarrow \text{FC}(64 \rightarrow C)$

Small RF · fast training · low parameter count

DeepCNN

$$RF = 1 + 3(k-1)$$

Three kernel-size-dependent convolutions with $32 \rightarrow 64 \rightarrow 128$ channels.

$\text{Conv}(k) \times 3 \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \times 2 \rightarrow \text{GAP} \rightarrow \text{FC}(128 \rightarrow 128) \rightarrow \text{FC} \rightarrow C$

Larger RF and capacity · highest accuracy · expensive for large kernels

DepthwiseCNN

$$RF = 1 + 3(k-1)$$

Three depthwise-separable blocks: depthwise spatial filtering + 1×1 pointwise mixing.

$[\text{DW}(k) + \text{PW}(1 \times 1)] \times 3 \rightarrow \text{MaxPool} \times 2 \rightarrow \text{GAP} \rightarrow \text{FC}(128 \rightarrow 128) \rightarrow \text{FC} \rightarrow C$

Same convolution-stack RF as DeepCNN · far fewer parameters for large kernels

Standard vs. Depthwise Convolution

SimpleCNN

$$RF = k$$

One standard convolutional layer.

$\text{Conv}(k) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \rightarrow \text{GAP} \rightarrow \text{FC}$

Standard convolution mixes spatial and channel information jointly.

DeepCNN

$$RF = 1 + 3(k-1)$$

Three standard convolutions.

$\text{Conv}(k) \times 3 \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \times 2 \rightarrow \text{GAP} \rightarrow \text{FC}$

Each filter spans all input channels, so cost grows rapidly with k^2 .

DepthwiseCNN

$$RF = 1 + 3(k-1)$$

Depthwise spatial filters followed by pointwise channel mixing.

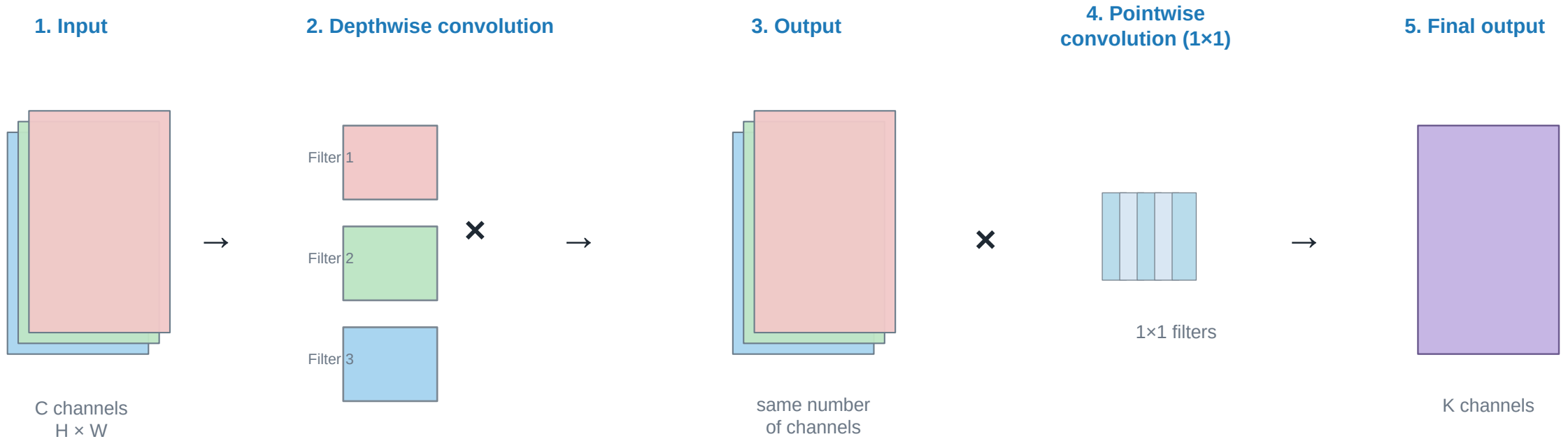
$[\text{DW}(k) + \text{PW}(1 \times 1)] \times 3 \rightarrow \text{MaxPool} \times 2 \rightarrow \text{GAP} \rightarrow \text{FC}$

Spatial filtering is channel-wise; 1×1 convolutions combine channels afterward.

Standard convolution learns spatial patterns and cross-channel relationships at the same time.

Each filter operates across all input channels simultaneously.

How Depthwise Convolution Works



Summary

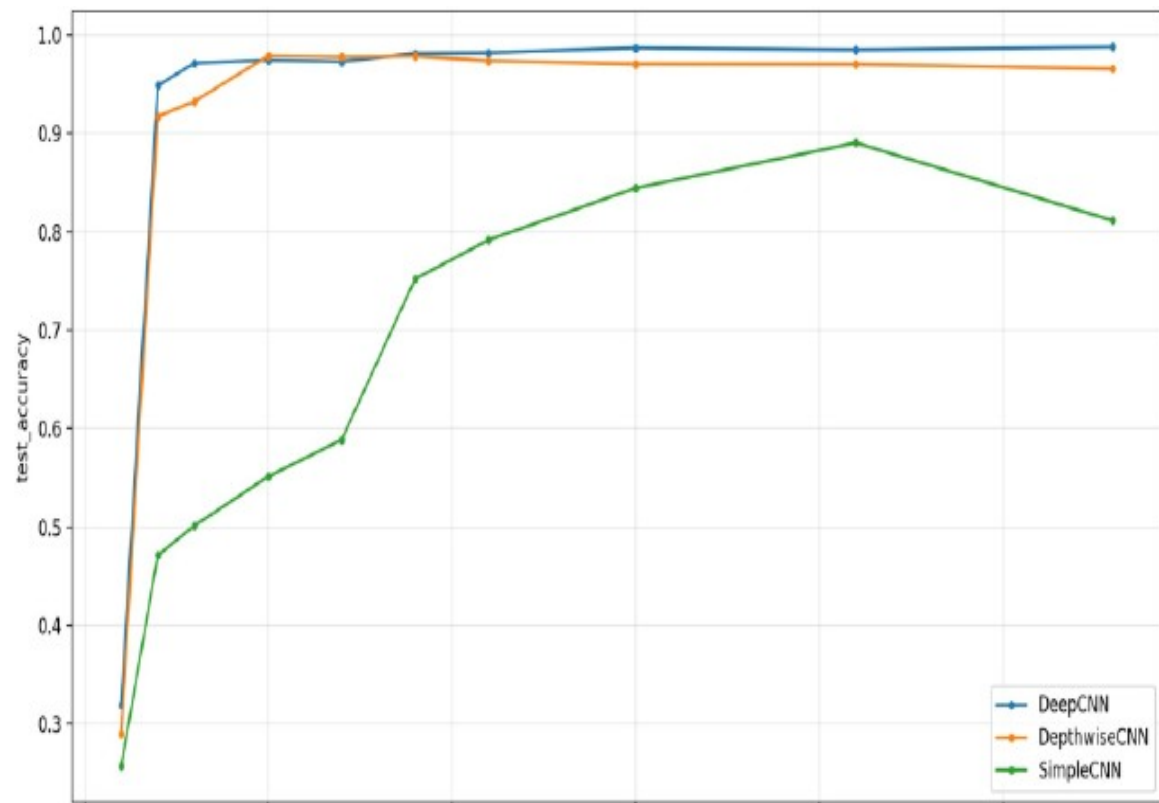
- Depthwise: one spatial 2D convolution per input channel
- Channels are not mixed during the spatial step
- Pointwise 1×1 convolution then combines channels
- The spatial receptive field is unchanged

Advantages

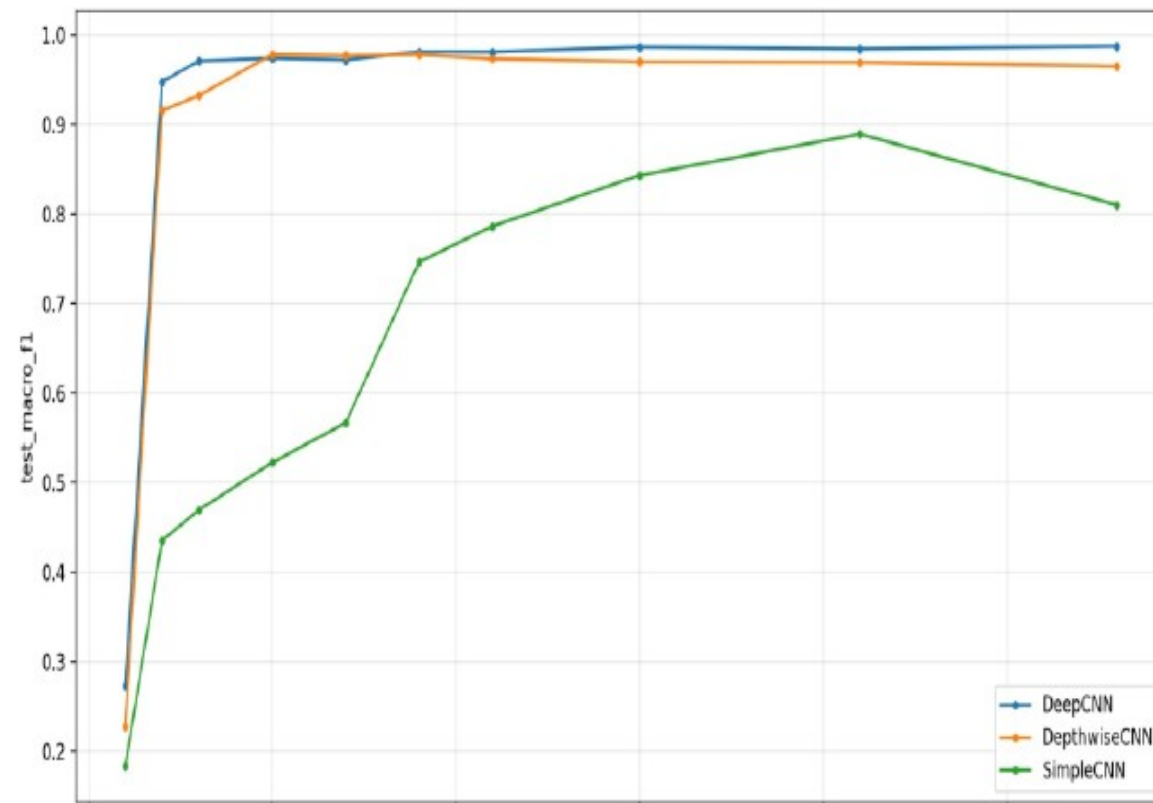
- Fewer parameters
- Lower computational cost (FLOPs)
- Faster, more efficient models
- Used in architectures such as MobileNet

MNIST Results - Test Accuracy & Macro-F1

Test Accuracy vs. Kernel Size



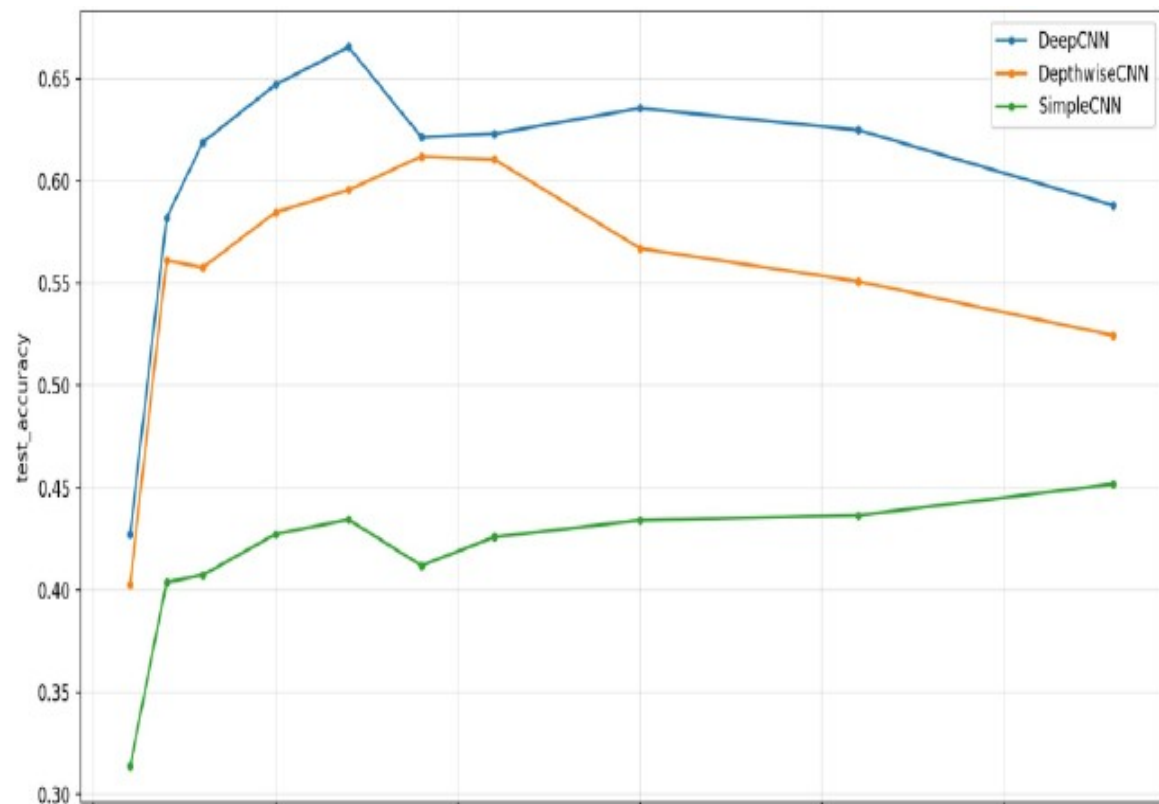
Test Macro-F1 vs. Kernel Size



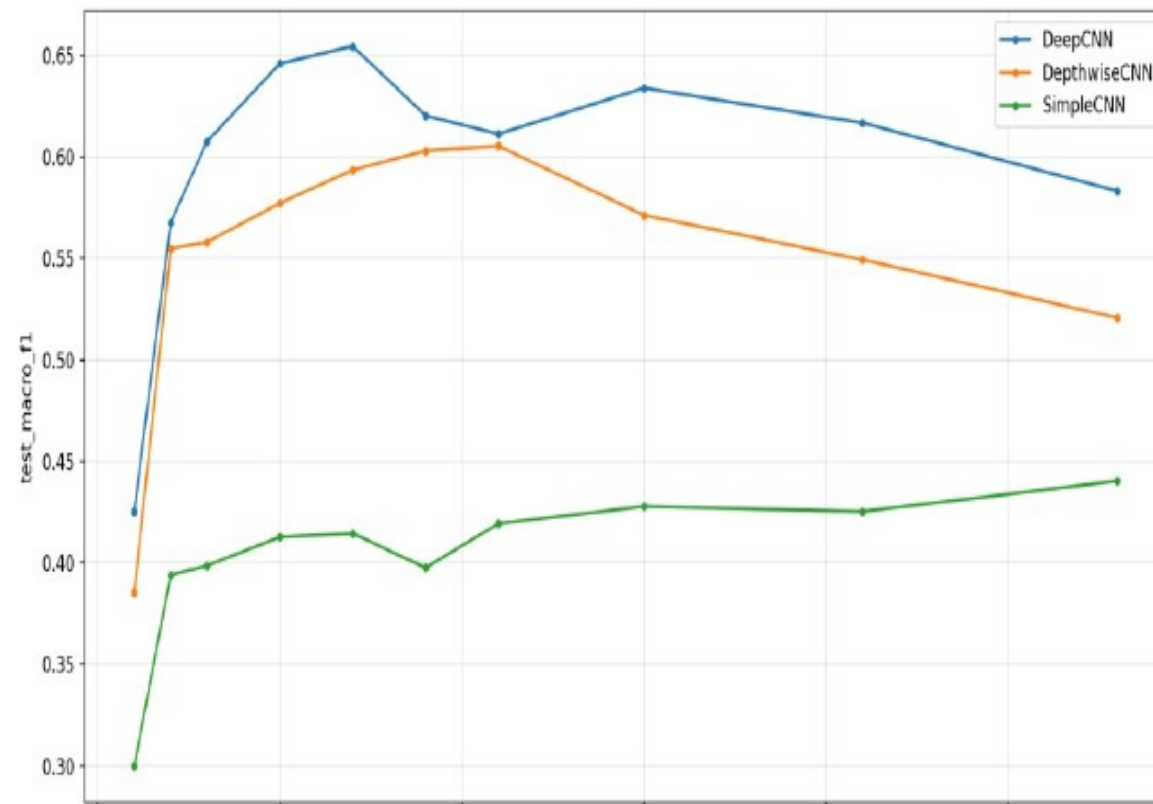
DeepCNN & DepthwiseCNN: ~97–99% from k=3 onward · SimpleCNN: 26% at k=1, peak ~89% at k=21 · Larger kernels partly compensate for a shallow model

CIFAR-10 Results - Test Accuracy & Macro-F1

Test Accuracy vs. Kernel Size



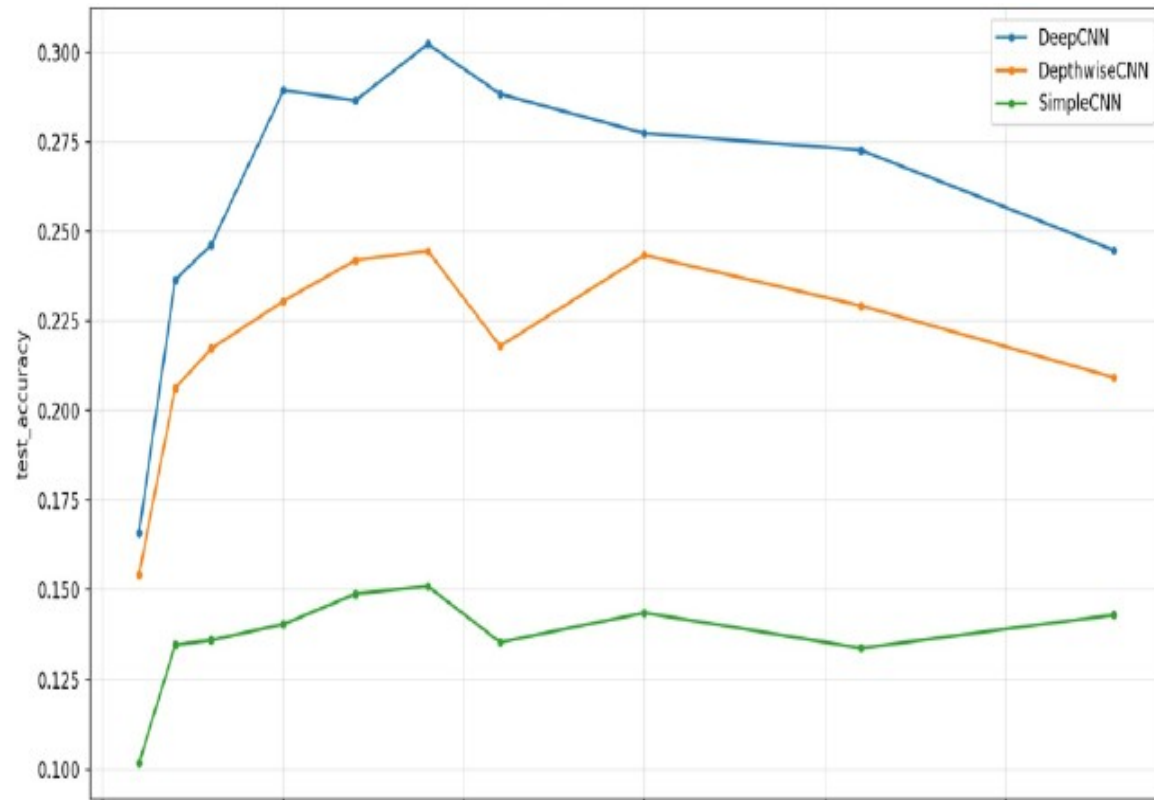
Test Macro-F1 vs. Kernel Size



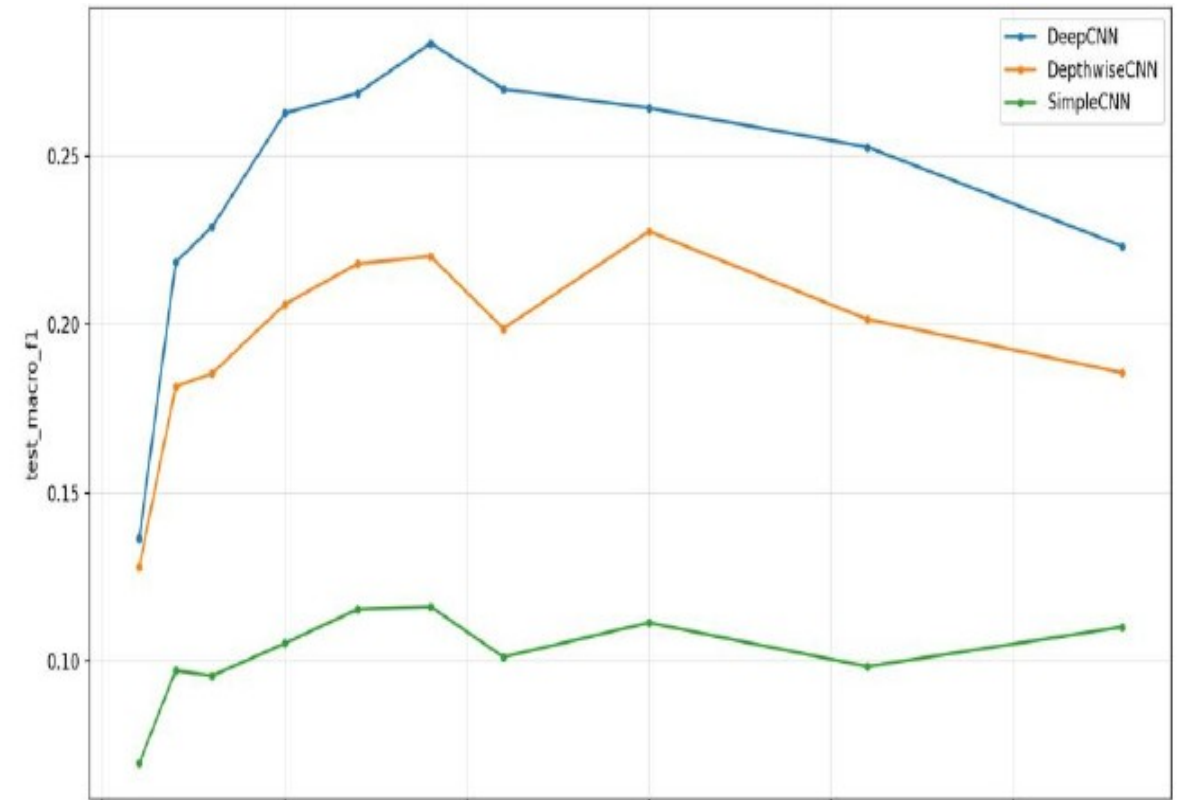
DeepCNN peaks at ~67% around k=7 · DepthwiseCNN reaches ~61% around k=9–11, then declines · SimpleCNN remains ~31–45%

CIFAR-100 Results - Test Accuracy & Macro-F1

Test Accuracy vs. Kernel Size



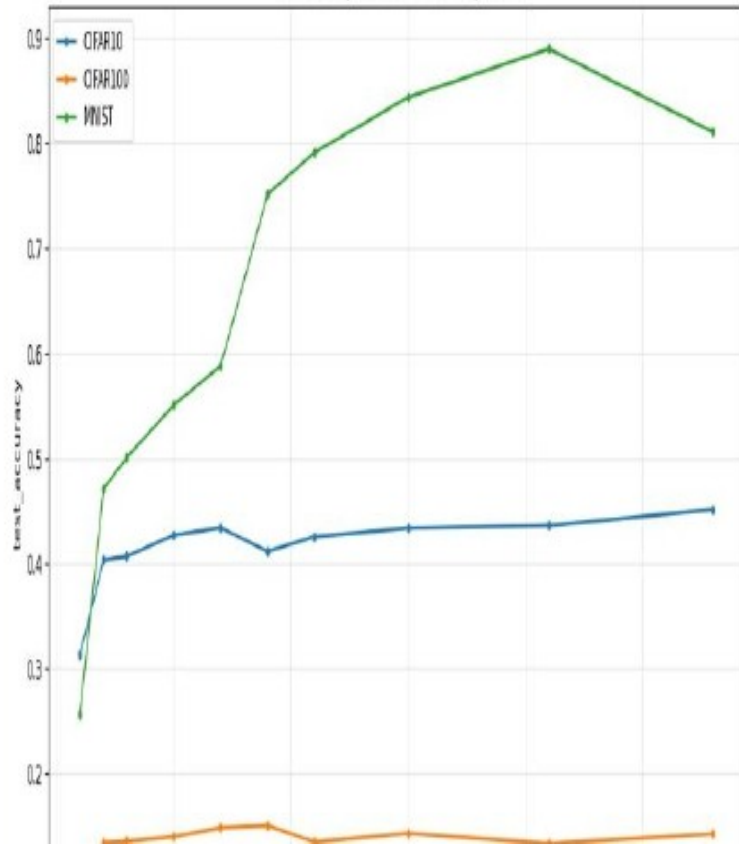
Test Macro-F1 vs. Kernel Size



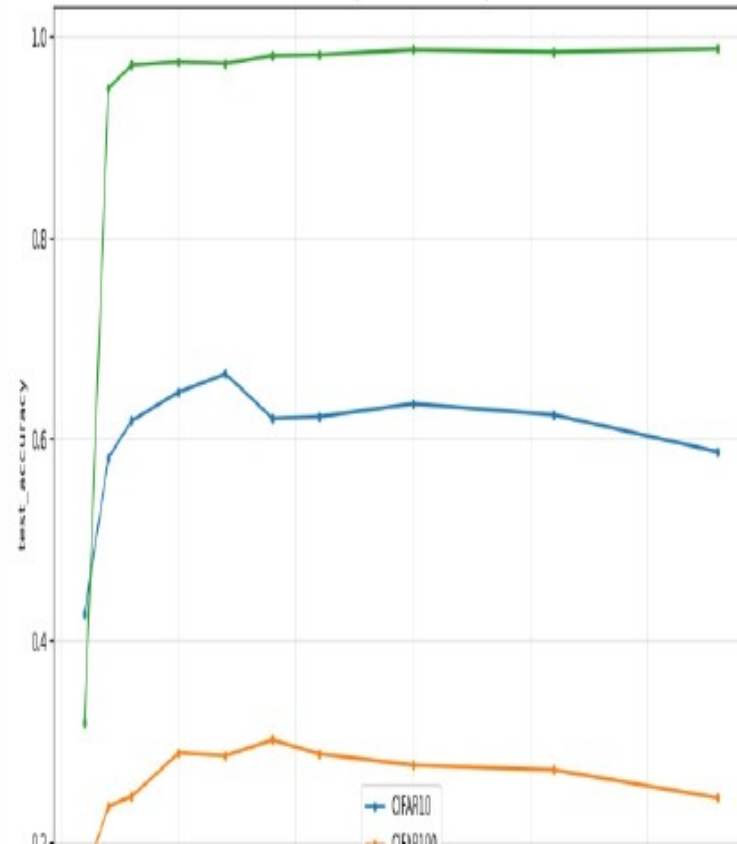
DeepCNN peaks at ~30% around k=9 · Performance drops for very large kernels · SimpleCNN stagnates around ~10–15%, indicating insufficient capacity for 100 classes

Comparison by Model Type Across Datasets

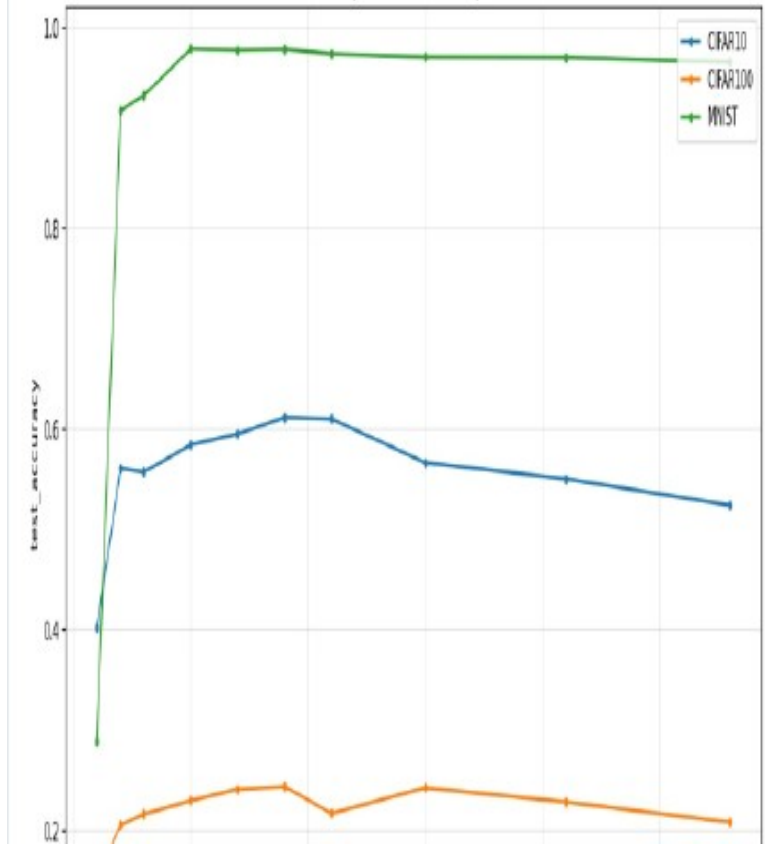
SimpleCNN



DeepCNN



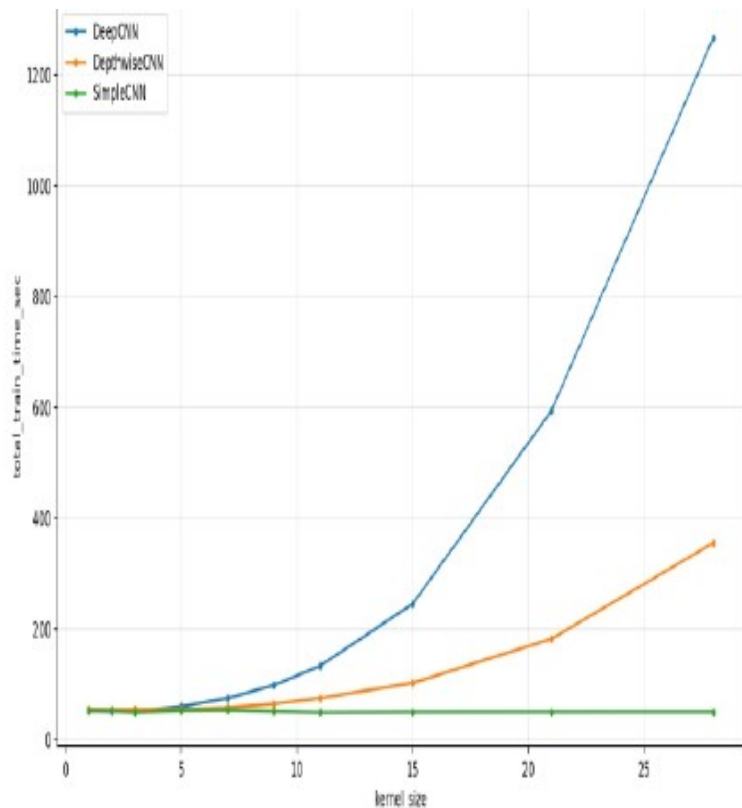
DepthwiseCNN



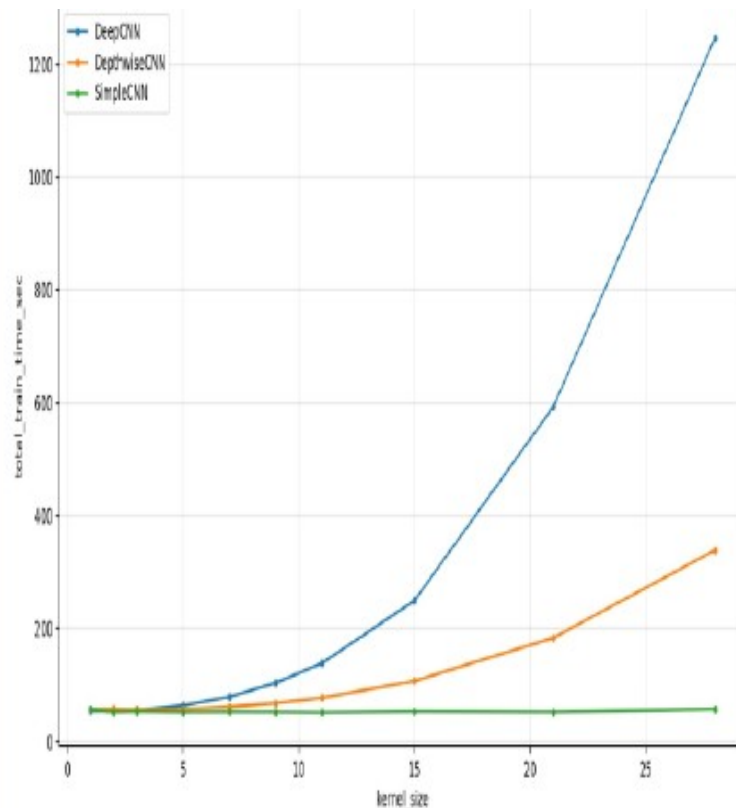
Across all architectures: MNIST is easiest, CIFAR-10 is intermediate, and CIFAR-100 is hardest because of the 100-class setting and small image size

Training Time vs. Kernel Size

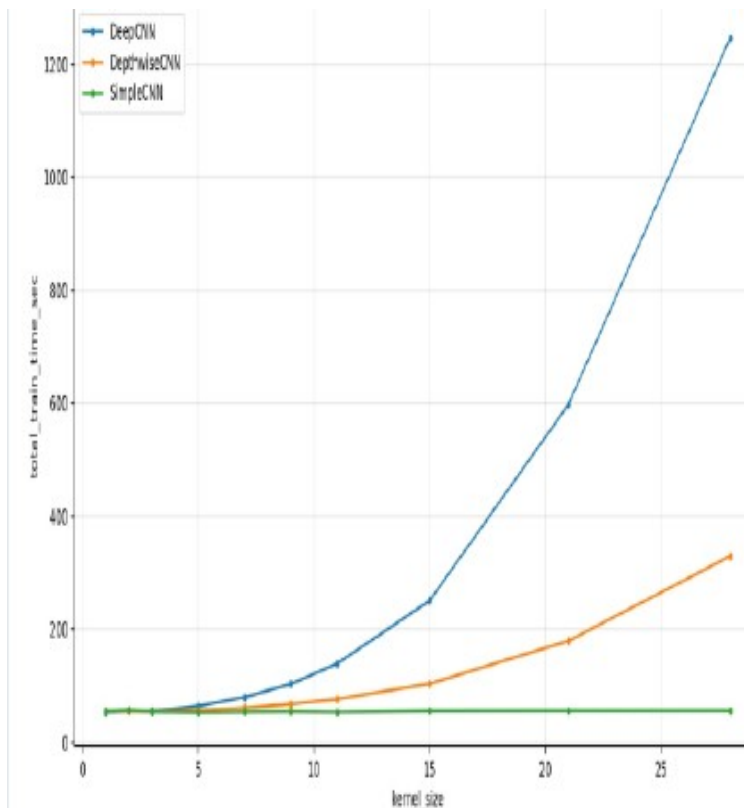
MNIST



CIFAR-10



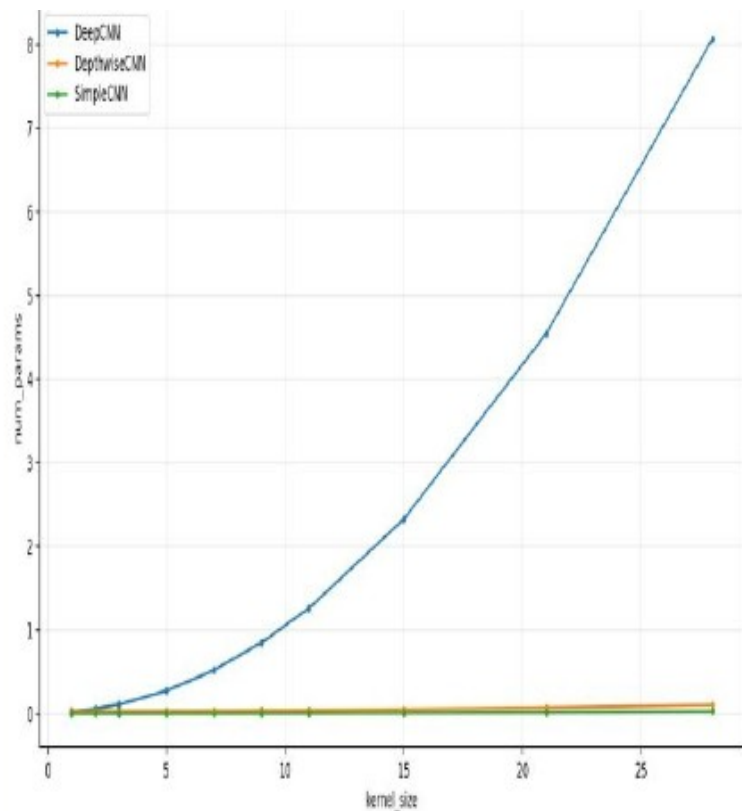
CIFAR-100



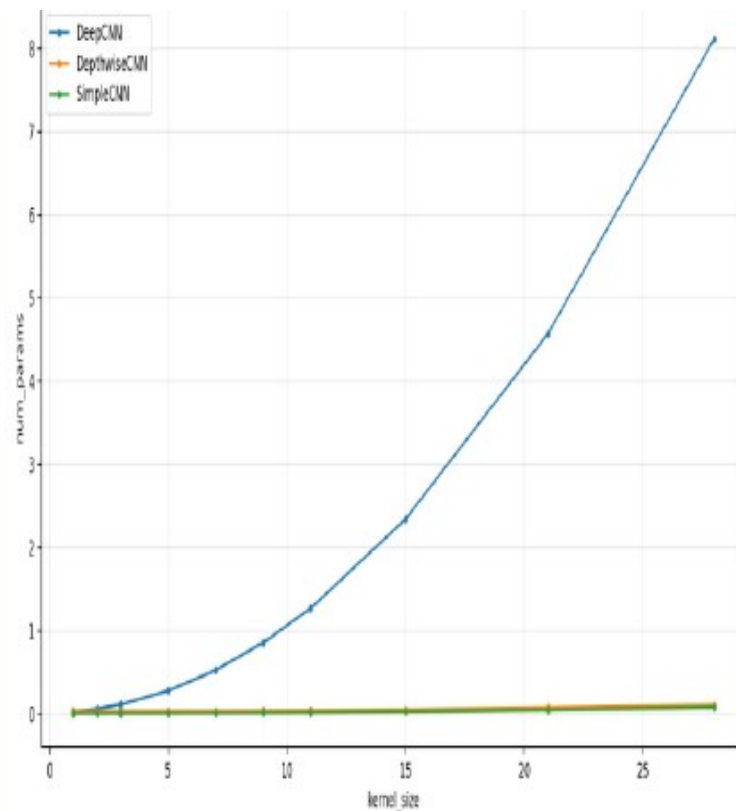
At k=28: DeepCNN ~1250 s (>20 min) · DepthwiseCNN ~350 s (~3.5× faster) · SimpleCNN ~50 s · Standard convolution cost grows roughly with k^2

Parameter Count vs. Kernel Size

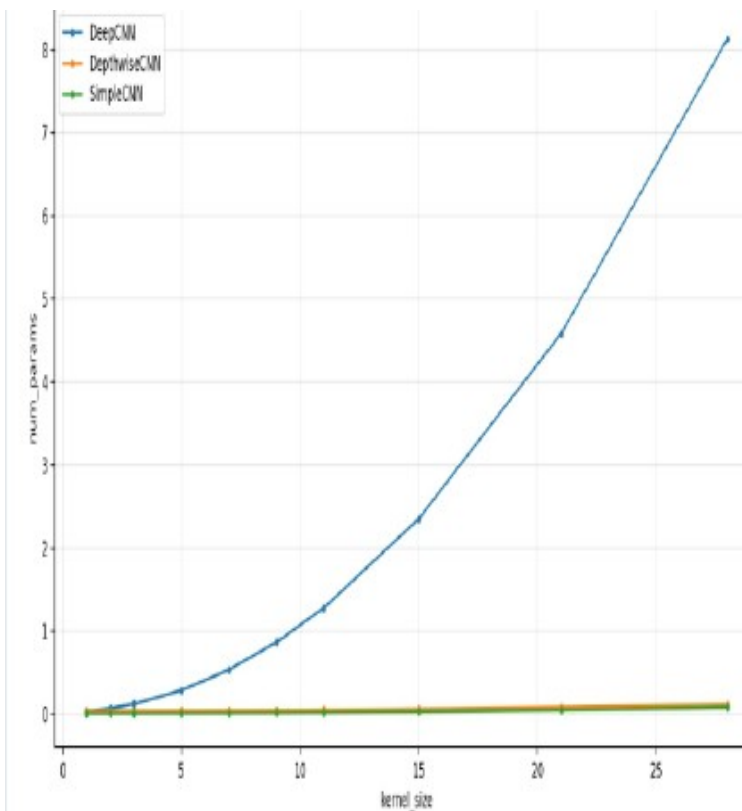
MNIST



CIFAR-10



CIFAR-100



At k=28: DeepCNN ~8M parameters · DepthwiseCNN ~130k (~60× smaller) · SimpleCNN ~90k · Separating depthwise and pointwise convolution dramatically reduces parameters

Global Average Pooling (GAP)

What does GAP do?

A pooling layer for CNNs

Averages each feature map

Produces one value per channel

Reduces the number of classifier parameters

```
self.classifier = nn.Sequential(  
    nn.AdaptiveAvgPool2d((1, 1)),  
    nn.Flatten(),  
    nn.Linear(128, 128),  
    nn.ReLU(inplace=True),  
    nn.Dropout(0.3),  
    nn.Linear(128, num_classes),  
)
```

Adaptive pooling automatically handles different spatial input sizes.



Summary & Conclusions

01

Optimal kernel size depends on the dataset

MNIST: $k=21$ for SimpleCNN, $k\approx 3-5$ for Deep/DW. CIFAR-10/100: $k\approx 7-9$ is usually strongest; very large kernels tend to hurt.

02

Depth matters more than kernel size for accuracy

DeepCNN consistently outperforms SimpleCNN at the same kernel size. More layers provide higher capacity and stronger feature extraction.

05

Receptive field $\neq$ performance

A larger receptive field is not automatically better: oversized kernels reduce performance for several model-dataset combinations.

03

DepthwiseCNN is an efficient compromise

Around $60\times$ fewer parameters and $\sim 3.5\times$ faster training than DeepCNN at $k=28$, with only a modest accuracy loss.

04

Training cost rises sharply with kernel size

DeepCNN grows from roughly ~ 50 s at small kernels to ~ 1250 s at $k=28$. Large standard kernels are computationally expensive.